

ECON 8310 Semester Project

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Executive Findings

- **Successful Automated Pipeline:** Processed 78 practice videos with consistent tracking results
- **Actionable Data Output:** Generated precise (x, y) coordinates for motion analysis
- **Strategic Metrics:** Enabled tracking for pitch trajectory, strike zone consistency, & launch angles
- **One-Click Replication:** Fully integrated GitHub and Google Colab workflow for immediate coaching use
- **IoU Score** of 0.0099

Data Challenges

CVAT was used to create training data

- Manual process of labeling the ball of interests in a box. CVAT allowed us to have a consistent format for training data

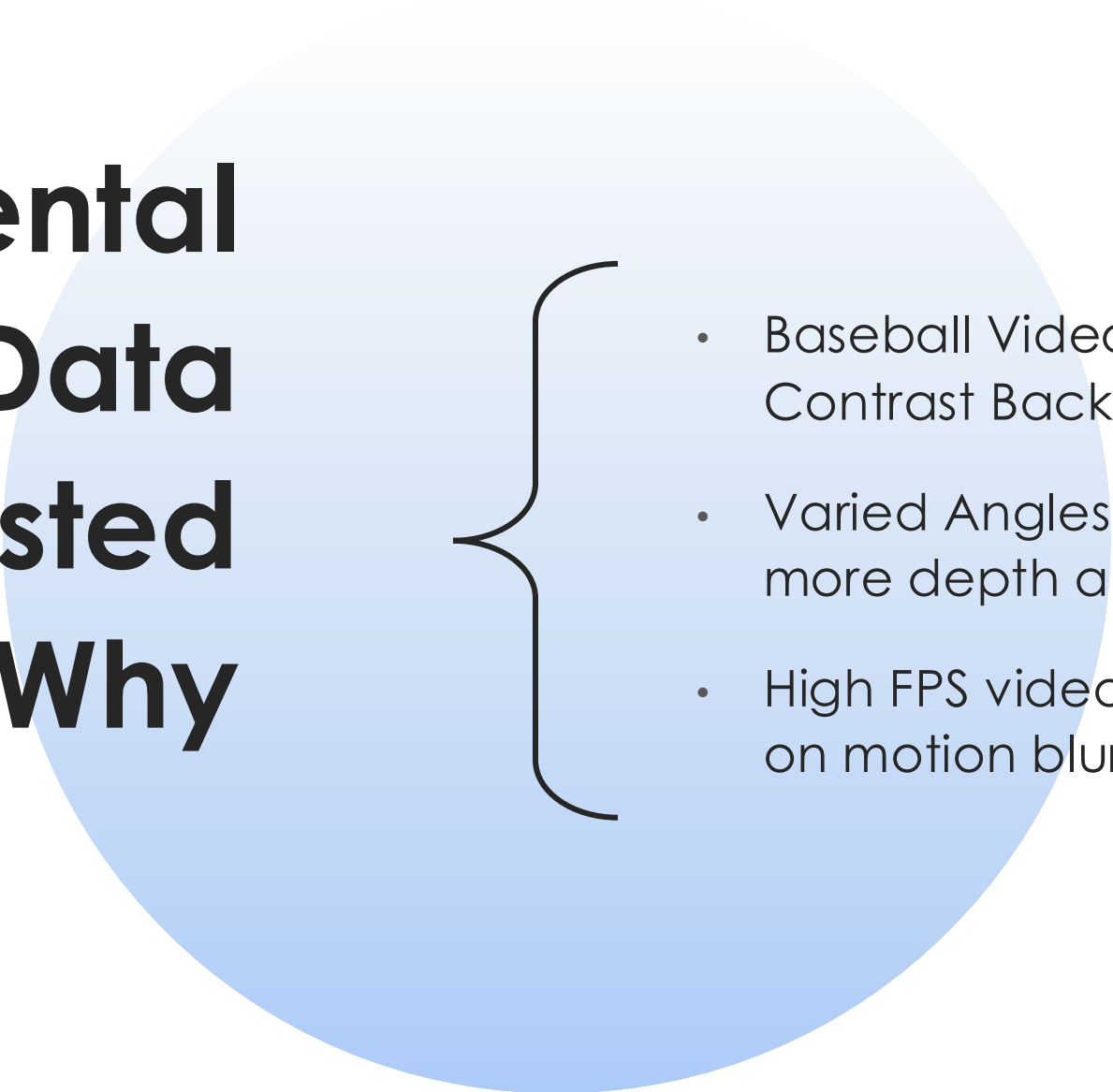
Challenges:

- Occlusions – View of ball being blocked
- Multiple Balls – Need to focus to be useful
- Light Backgrounds and "Noise" Objects

Training Decisions

- Only focus on "ball of interest" and ignored any other balls in the background
- Prevents model from getting distracted and greater use in coaching capacity

Supplemental Data Requested and Why

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- Baseball Videos with a High Contrast Background
 - Varied Angles to understand more depth and scale
 - High FPS videos to better train on motion blur

Frame Differencing (Isolating Motion)

- **The "Static" Problem:** Training environments (nets, cages, turf) create background "noise"
- **Motion Isolation:** Frame Differencing subtracts the background to highlight only moving pixels
- **Efficiency:** Reduces the "search area" for the neural network
- **Result:** Faster identification and fewer "false positives" from stationary objects



Neural Network Architecture

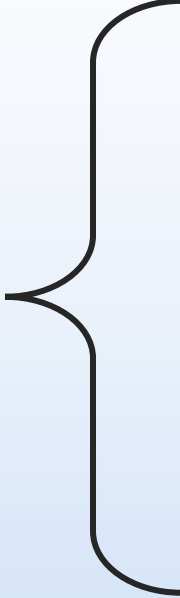
- **Transfer Learning:** Built on a ResNet-18 base model
- **Foundational Knowledge:** Pre-trained on millions of images to recognize basic shapes and edges
- **Custom Fine-Tuning:** Trained specifically on our labeled baseball dataset
- **Regression Pivot:** Re-wired the final layer to output 4 specific coordinates
 - (x, y, width, height).

Measuring Success (IoU)

- **Intersection over Union (IoU):** The gold standard for object detection accuracy.
- **The "Tightness" Metric:** Measures overlap between our label and the model's prediction.
- **Goal:** Maximize IoU toward 1.0 for professional-grade diagnostics.
 - **Reality:** Highest accuracy achieved was 0.0099



Conclusion & Implementation

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- **Summary:** Automated tracking turns video into a strategic coaching asset
 - **Replication:** Fully reproducible via GitHub and Google Colab
 - **Future Scope:** Multiple ways to improve model so it's more grounded in reality